Programming Project

(20 points)

The use of computers in education is referred to as computer-assisted instruction (CAI). Write a program that will help an elementary school student learn multiplication. Use the rand function to produce two positive one-digit integers. The program should then prompt the user with a question, such as

How much is 6 times 7?

The student then inputs the answer. Next, the program checks the student's answer. If it's correct, display one of the following messages **randomly**;

- Very good!
- Excellent!
- Nice work!
- Keep up the good work!

and ask another multiplication question. If the answer is wrong, display one of the following messages **randomly**;

- No. Please try again.
- Wrong. Try once more.
- Don't give up!
- No. Keep trying.

and let the student try another question from the same level repeatedly until the student finally gets it right 3 times in a row. A separate function should be used to generate each new question. This function should be called once when the application begins execution and each time the user answers the question correctly.

Your program should count the number of correct and incorrect responses typed by the student. After the student types 20 answers, your program should calculate the percentage that are correct. If the percentage is lower than 75%, display " Please ask your teacher for extra help .", then reset the program for another trial. If the percentage is 75% or higher, display " Congratulations, you are ready to go to the next level! ", then reset the program for another trial.

The program should allow the user to start with questions of low difficulty level. The difficulty level should move to the medium level when the student manages to answer 3 questions correctly in a row. Finally, the student moves to the hard level when she/he answers 3 questions correctly in a row.

The program should use only single-digit numbers in the low difficulty level problems; at the medium difficulty level problems, numbers as large as two digits, and 3 digit numbers for the hard level problems.

Your home screen should be like the following example;

```
        Educational GAME


____________________________________________
            WELCOME
              to
          THE EDU GAME
____________________________________________
____________________________________________
   Do you want to master math?!
____________________________________________
____________________________________________
 > Press S to start the game
 > Press V to view the highest score
 > press H for help
 > press Q to quit
____________________________________________

Please enter your choice here:
```

If the student starts the game, first she/he must be asked for her/his name:

```
*********************************************************

        Please enter your username:Abdallah


*********************************************************
```

Then, the questions may be displayed as following;

```
How much is 5 times 9?

A.5             B.59

C.54            D.45
____________________________________________

Please enter your choice here:
```

The high score would be displayed as following;

```
************************************************************

         Abdallah has the Highest Score 18.70

************************************************************
```

Bonus Question (Extra 5 Points)

The program may ask the student questions from a different field, i. e. Science, history, English or geography. In addition to that, the questions are grouped in 3 files; one for each difficulty level. The program then chooses the question from the files randomly.

Project Guidelines

- Your program must contain at least the following functions:
  - home() - show the home screen menu
  - edit_score() - edit the user score with each correct answer
  - help() – help menu with game summary and rules
  - show_score() – to view the highest score
- The students may be grouped in teams of maximum 6 members.
- The project delivery would be on Sunday 25th of December.

Useful functions you may use

rand(): a function to generate random numbers. The function needs the library stdlib.h to be included. Before using the function you need to initialize it as follows:

```
time_t t;
srand((unsigned) time(&t));
```

Then, each time you need to generate a random number for instance a number between 0 to 10 you may call the function as follows:

```
int n = rand() % 11;
```

system(“cls”): A function to clear the screen on Windows based machines. The function needs the library stdlib.h to be included. If you run your code on a Linux based machine, then you need to change the function call to be system(“clear”).